Homework #1 — Morbidity & Mortality (Chapters 3–4)

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Part 1

A. Numerators and Denominators

For each quantity below, write the numerator, denominator, and time window.

(i)

Incidence rate of RSV in person-years for children under 5 in Weber County in 2022

- **Numerator:** Number of new cases of RSV
- **Denominator:** Population of children under 5
- **Time Window:** The year 2022

(ii)

Point prevalence of asthma on Jan 1, 2026 in a panel of pediatric patients at Utah Valley Regional Medical Center.

- **Numerator:** Number of current asthma cases
- **Denominator:** Number of pediatric patients that we have
- **Time window:** That specific day

(iii)

Average cause-specific mortality rate for heart disease in Florida in 2021-2025

- **Numerator:** Deaths from specifically heart disease
- **Denominator:** Population of Florida in the middle of 2023
- **Time window:** From 2021-2025

B. Fill in the blanks:

- Incidence counts **new** cases over a **fixed** period.
- Prevalence counts **active** cases at a **point in time** (or during a period).
- Mortality counts **deaths** over a **fixed** period.

C. Short Responses

(i) Give one reason prevalence could increase even if incidence stays the same.

Prevalence would increase if the window of time in which someone has the disease is extended.

(ii) Give one reason measured incidence could increase even if true incidence stays the same.

There could be changes to diagnostic criteria that make it so more people are diagnosed with the disease.

Part 2 — Computing Morbidity and Mortality Measures (by hand)

Show all work. Clearly label units.

A. Cumulative Incidence

A cohort of $N = 2,500$ individuals is disease-free at baseline. Over a follow-up period of $T = 3$ years, $C = 75$ new cases of Disease X occur. Assume complete follow-up.

(i) Formula and Calculation

$$\text{Cumulative Incidence} = \frac{\text{Number of new cases during the period}}{\text{Number at risk at the start of the period}} = 75/2500 = 0.03$$

(ii) Per 100,000

$$\text{CI per } 100,000 = 0.03 \times 100,000 = 3,000$$

B. Incidence Rate Using Person-Time

A study follows individuals until disease or censoring:

- 10 people followed for 2.0 years
- 8 people followed for 1.5 years
- 12 people followed for 0.5 years
- Total incident cases observed: 9

(i) Compute total person-time

$$\text{Person-years} = (10)(2) + (8)(1.5) + (12)(0.5) = 38 \text{ units}$$

(ii) Compute incidence rate per 100 person-years

$$\text{Incidence Rate} = \frac{9}{38} = 23.7 \text{ units}$$

C. Point vs. Period Prevalence

A clinic follows $N = 800$ patients.

- On January 1, 2026, 40 patients have asthma
- During 2026, 90 patients had asthma at any point

(i) Compute Point Prevalence

$$\text{Point Prevalence} = \frac{40}{800} = 5 \text{ per hundred}$$

(ii) Compute Period Prevalence

$$\text{Period Prevalence} = \frac{90}{800} = 11 \text{ per hundred}$$

(iii) Fill in

Period prevalence must be **greater** than point prevalence.

D. Mortality Rate vs. Proportionate Mortality

Two communities:

Community	Population	Total deaths	Heart disease deaths
A	10,000	200	60
B	10,000	100	30

(i) All-cause mortality rate per 1,000

Community A:

$$\frac{200}{10,000} \times 1,000 = 20 \text{ per } 1000$$

Community B:

$$\frac{100}{10,000} \times 1,000 = 10 \text{ per } 1000$$

(ii) Proportionate mortality

Community A:

$$\frac{60}{200} = 30\%$$

Community B:

$$\frac{30}{100} = 30\%$$

(iii) Conceptual question

Explain why proportionate mortality does not measure the risk of dying from heart disease:

Proportionate mortality is only the measure of, given you died, the chances of your death being of heart disease. If a community has a higher chance of dying overall the average person would have a higher chance of dying from heart disease, even if the proportionate mortality of heart disease compared to all other causes remains the same.

3. Using R with Lyme disease morbidity calculations

A. Setup

```r
library(dplyr)
library(readr)
library(tidyr)
library(ggplot2)
library(sf)
library(usmap)

# Load the dataset
county_data <- read_csv("lyme_data.csv")

# Quick overview of column names
names(county_data)
```

```
 [1] "Ctyname"   "stname"    "ststatus"  "stcode"    "ctycode"   "Cases2001"
 [7] "Cases2002" "Cases2003" "Cases2004" "Cases2005" "Cases2006" "Cases2007"
[13] "Cases2008" "Cases2009" "Cases2010" "Cases2011" "Cases2012" "Cases2013"
[19] "Cases2014" "Cases2015" "Cases2016" "Cases2017" "Cases2018" "Cases2019"
[25] "Cases2020" "Cases2021" "Cases2022" "TOT_POP"
```

B. County-level cumulative incidence (2018–2022) and map it.

Goal: For each county, compute the 5-year cumulative incidence:

$CI_{2018-2022} = \frac{\text{Cases2018+Cases2019+Cases2020+Cases2021+Cases2022}}{\text{TOT\_POP}}$

Then plot it on a US county map. Use code from January 13th lecture.

(i)

Compute the indicence rate.

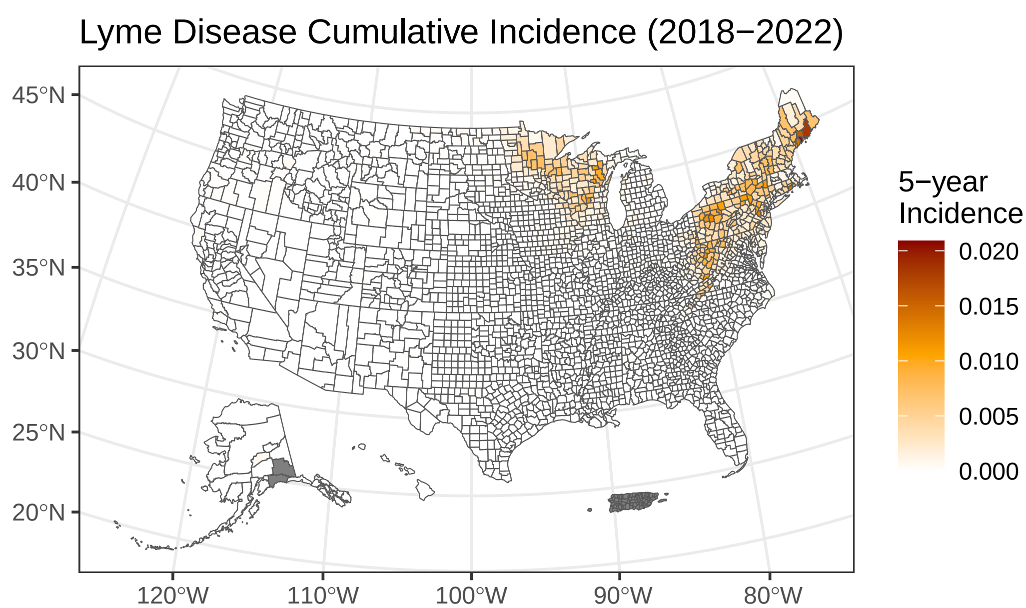
```r
# Requirements:
# - Store the resulting dataset as county_ci
# - Fill in the commented code.
county_ci <- county_data |>
  mutate(
```

```
    cases_5yr = Cases2018 + Cases2019 + Cases2020 + Cases2021 + Cases2022,
    ci_2018_2022 = cases_5yr / TOT_POP,
    FIPS = paste0(sprintf("%02d", as.numeric(stcode)),
                  sprintf("%03d", as.numeric(ctycode)))
  )
```

(ii)

Plot the incidence rate. Please update the orange and darkred in the color scheme.

```
usa_counties <- us_map("counties") %>%
  left_join(county_ci, by = c("fips" = "FIPS"))

ggplot(usa_counties) +
  theme_bw() +
  geom_sf(aes(fill = ci_2018_2022)) +
  scale_fill_gradientn(colors = c("white", "orange", "darkred")) +
  labs(title = "Lyme Disease Cumulative Incidence (2018-2022)",
       fill = "5-year\nIncidence")
```



C. Maine state-level rates by year (sum over counties)

Goal: For each year 2001–2022:

1. Sum cases across all Maine counties
2. Sum population across all Maine counties
3. Compute the state-level rate per 100,000
4. Print the first 10 rows of the `maine_state` data frame
5. Plot the state-level rate over time

(i) Compute and output table

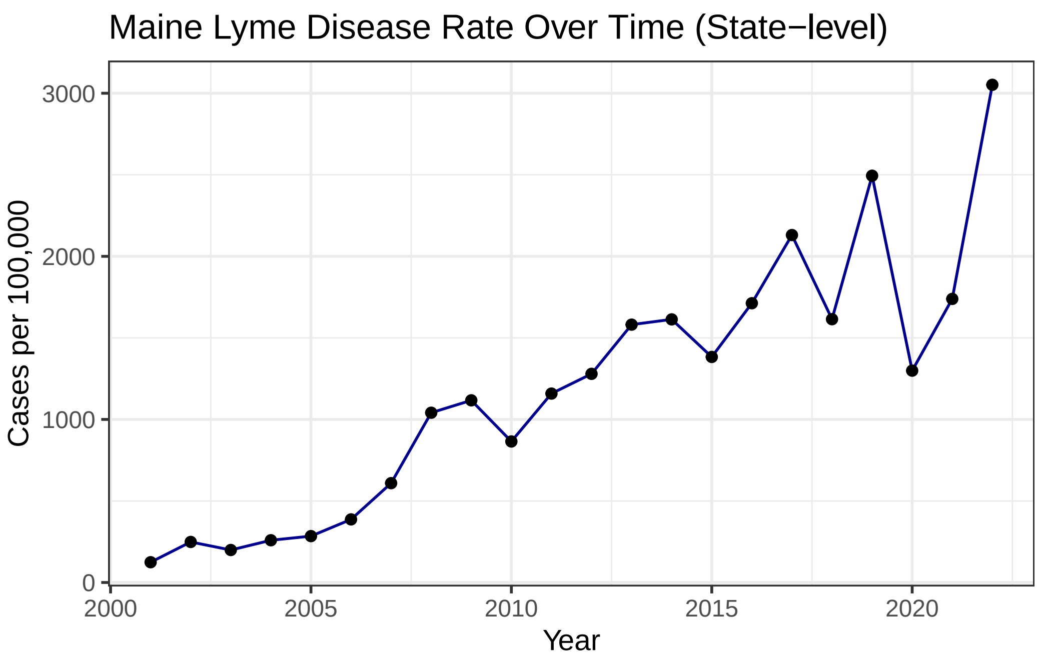
```r
# Start with the code below
# Hint: use group_by(year) and summarise() to aggregate county data.
# create variable rate_per_100k

maine_state <- county_data %>%
  filter(stname == "Maine") %>%
  pivot_longer(cols = starts_with("Cases"), names_to = "year", values_to = "cases") %>%
  mutate(year = as.integer(gsub("Cases", "", year))) %>%
  filter(year >= 2001 & year <= 2022) %>%
  group_by(year) %>%
  summarise(case_total = sum(cases), population = mean(TOT_POP)) %>%
  mutate(
    rate_per_100k = (case_total / population) * 100000
  )

slice_head(maine_state,n = 10)
```

```
# A tibble: 10 x 4
    year case_total population rate_per_100k
   <int>      <dbl>      <dbl>         <dbl>
 1  2001        108     86834.          124.
 2  2002        216     86834.          249.
 3  2003        173     86834.          199.
 4  2004        225     86834.          259.
 5  2005        247     86834.          284.
 6  2006        336     86834.          387.
 7  2007        529     86834.          609.
 8  2008        904     86834.         1041.
 9  2009        970     86834.         1117.
```

```
10  2010         751      86834.           865.
```

(ii) Plot Maine's state-level rate over time

```
ggplot(maine_state, aes(x = year, y = rate_per_100k)) +
  geom_line(col = "darkblue") +
  geom_point() +
  theme_bw() +
  labs(title = "Maine Lyme Disease Rate Over Time (State-level)",
       x = "Year", y = "Cases per 100,000")
```



(iii) Highest and Lowest Rates

Write your answers here:

- Highest rate year (2001–2022) and that year's rate: 2022 had a rate of 3052 per 100,000
- Lowest rate year (2001–2022) and that year's rate: 2001 had a rate of 124 per 100,000